

SOE 411

Open-Source Software Development and Practice

OPEN-SOURCE LICENSING

INTRODUCTION

- ✓ Open source software is that computer software for which the source code is made available to the public in such a way that anyone can view, use, modify, and distribute the software.
- ✓ It is typically developed collaboratively by a community of volunteers or organizations with the common goal of creating high-quality, accessible software for everyone.
- ✓ Open source software has undoubtedly become a powerful force in the software industry, enabling collaborative development, reducing costs, fostering innovation, and providing users with a wide range of high-quality software options across various domains, from operating systems (e.g., Linux) to web development frameworks (e.g., Apache) and office productivity suites (e.g., LibreOffice).

How do we know one? - Users have:

1. **Access to Source Code:** The users can access the source code of the software, which is the human-readable such that programmers can understand how it works by scanning through the program codes.
2. **Permission to Modify:** Since users are allowed access to the sources codes, they are also permitted to modify code to meet their peculiar needs or to fix bugs and security vulnerabilities they have found.
3. **Distribution Rights:** Users are often allowed to distribute copies of the software to others, whether for free or for a fee, as long as they comply with the terms of the open source license.

How do we know one? - Users have:....

1. Equal access and rights for all users irrespective of race.
2. No Restriction on Usage. It can be used for any purpose, including commercial and non-commercial use, without restrictions imposed by the software's creators.
3. Freedom of usage. Open source licenses, such as the GNU General Public License (GPL) or the MIT License, are designed to protect these freedoms and ensure that they are passed on to subsequent users and developers.

Are Open Source Software Licensed?

1. The answer is a big ‘Yes’. There are different types of licensing available in the open source domain.

But, What is software licensing?

1. Software licensing refers to the legal mechanism that governs how software or other creative works can be used, shared, and distributed.
2. It establishes the terms and conditions under which individuals or organizations are granted permission to use the software while still respecting the rights of the software's creator or owner.

Is software licensing necessary?

Since a software is an intellectual property, licensing plays a crucial role of defining the terms and conditions under which certain rights and permissions are granted or restricted to or from the user. Some importance and benefits of licensing are listed below:

- **Legal Framework:** Licensing establishes a legal framework or guidelines that governs the use, distribution, and protection of intellectual properties both licensors and licensees.
- **Protection of Intellectual Property:** Licensing allows creators and owners of intellectual property to protect their work by specifying how others can use it. It helps prevent unauthorized use, copying, or distribution, reducing the risk of intellectual property theft or infringement.

Some importance and benefits of licensing.

- **Monetization:** Licensing provides a means for creators and copyright holders to monetize their intellectual property. They can grant licenses in exchange for fees, royalties, or other forms of compensation, hence, generating income from their creations.
- **Market Expansion:** Licensing can help creators reach broader markets and audiences by allowing third parties to use, sell, or distribute their products or content. This can lead to increased visibility and sales of such intellectual properties.
- **Collaboration:** Licensing encourages collaboration and partnerships among individuals, organizations, and industries. It enables the sharing of knowledge, technology, and resources while defining the terms of such cooperation.

Some importance and benefits of licensing..

- **Innovation:** Open licensing models, such as open source software licenses, foster innovation by making source code or intellectual property accessible to a wider community of developers and contributors. This can lead to rapid development and improvement.
- **Compliance and Accountability:** Licensing ensures that both licensors and licensees adhere to agreed-upon terms and conditions. It establishes accountability and legal standpoint in case of disputes or violations.
- **Standardization:** In industries where interoperability and compatibility are essential (e.g., technology and telecommunications), licensing can define standards and specifications that help ensure products work together seamlessly. It defines the limit of use.

Some importance and benefits of licensing...

- **Consumer Protection:** Licensing can protect consumers by setting quality standards and ensuring that products or services meet certain criteria. It provides transparency regarding the terms of use and warranty.
- **Ethical Considerations:** Licensing can incorporate ethical considerations into the use of intellectual property. For example, open access licenses for academic research promote the sharing of knowledge for the public good.
- **Flexibility:** Licensing allows creators and rights holders to tailor permissions and restrictions to suit their objectives. They can choose from a range of license types, such as open source licenses, proprietary licenses, and creative commons licenses, to best meet their needs.
- **Global Reach:** Licensing facilitates international distribution and collaboration by addressing legal and jurisdictional issues, making it possible to share and protect intellectual property across borders.

Types of Software Licenses

- There are several types of software licenses, each with its own set of terms and conditions that govern how the software can be used, modified, and distributed.
- The choice of a specific type of software license largely depends on the intentions of the developer.
- These license types will be discussed in detail: Proprietary (Closed Source) License, Open Source License, Permissive License, Copyleft License, Commercial License, Public Domain License, Dual License

Proprietary (Closed Source) License:

- Restrict access to the source code of the software.
- There are restrictions on *modification, redistribution, and reverse engineering*.
- Often associated with commercial software products.
- **Examples:** Microsoft Windows, Adobe Photoshop, and many commercial software applications.

Open Source License

- Grant users various freedoms to view, use, modify, and distribute the software's source code.
- Promote transparency and collaboration.
- Users have access to the source code, freedom to modify and distribute, with varying degrees of restrictions (permissive or copyleft).
- Users are encouraged to join community collaborations.
- **Examples:** GNU General Public License (GPL), MIT License, Apache License, and more.

Permissive License

- Permissive licenses, also known as ***non-copyleft*** licenses, allow users to modify and distribute the software under minimal restrictions.
- Are less restrictive than copyleft licenses.
- With this license, users have freedom to use, modify, and redistribute with minimal requirements.
- This may only require attribution or inclusion of the original license. **Examples:** MIT License, BSD Licenses (2-Clause and 3-Clause), and the Apache License.

Copyleft License

- Are a subset of open source licenses that *enforce the requirement that derivative works be licensed under the same terms as the original work.*
- Aim to ensure that the software remains open source. In other words, you don't take what is free to make what you will sell.
- Enforces the share-alike principle. This means that every derivative works must use the same license.
- Ensures strong commitment to open source principles.
- **Examples:** GNU General Public License (GPL), GNU Lesser General Public License (LGPL).

Public domain licenses

- Public domain licenses allow creators to relinquish their rights over the software, effectively placing it in the public domain.
- Users can use, modify, and distribute without any restrictions.
- It presents no copyright or ownership claims, hence providing maximum freedom for users.
- Examples: Creative Commons CC0.

Dual licenses

- Dual licensing involves offering the software under two different licenses, typically a free/open source license **and** a commercial license.
- Users can choose the license that best suits their needs.
- Offers flexibility for users to choose the license, and is often used for commercial viability while maintaining an open source option.
- Examples: MySQL (offered under both GPL and commercial licenses).

Understanding Software license types

- The choice of license depends on the following:
 - The goals of the software developer or organization, and
 - The desired level of openness, and the intended audience.
- It is very important to carefully read and understand the terms and conditions of a license before using or distributing software. This section provides a brief overview of some license types.

A. GNU General Public License (GPL)

- The GNU General Public License (GPL) is a widely used *copyleft license* created by the Free Software Foundation (FSF) which ensures that derivative works of GPL-licensed software *remain* open source.
- The following are the key characteristics of the software license:
 - i. Requires that any derivative works also be licensed under the GPL.
 - ii. Strong emphasis on user freedoms and the right to access and modify source code.
 - iii. Encourages contributions to the open source community.
- GPL is commonly used for applications, libraries, and software projects that prioritize keeping software *open source*, such as the Linux operating system and the GNU tools.

B. MIT License

- The MIT License is a permissive open source license that allows users to use, modify, and distribute the software with minimal restrictions. It is one of the most permissive open source licenses. The following are the key characteristics of the software license:
 - i. Requires inclusion of the original copyright notice and disclaimer.
 - ii. Allows for both personal and commercial use of the software.
 - iii. Minimal obligations, providing maximum freedom to users.
- The MIT License is popular for open source libraries, frameworks, and smaller projects due to its permissive nature.

C. Apache License

- The Apache License is an open source license created by the Apache Software Foundation. It is a permissive license that combines open source principles with protections against patent litigation. The following are the key characteristics of the Apache license:
 - i. Permits usage, modification, and distribution with proper attribution and notices.
 - ii. Offers patent grant provisions to protect users from patent-related claims.
 - iii. Requires licensees to state changes made to the software when redistributing.
- The Apache License is commonly used for projects related to web servers, cloud computing, and big data processing, including the Apache HTTP Server and Apache Hadoop.

D. BSD Licenses (e.g., 2-Clause and 3-Clause BSD)

- BSD (Berkeley Software Distribution) Licenses are a family of permissive open source licenses known for their minimal restrictions. The 2-Clause and 3-Clause BSD licenses are variations with slightly different terms. The following are the key characteristics of the BSD license:
 - i. Allow usage, modification, and distribution with minimal requirements.
 - ii. Require attribution and retention of copyright notices in the original software.
 - iii. Do not include a requirement to include the license text in derivative works (2-Clause).
- BSD licenses are commonly used for both open source and proprietary software, particularly in academia and networking (e.g., FreeBSD and OpenBSD).

E. Creative Commons Licenses (for non-software works):

- Creative Commons licenses are not specific to software but are used for various creative works, including art, music, literature, and educational content. These licenses allow creators to define the terms of use for their works. The following are the key characteristics of the Creative Commons license:
 - i. Multiple license options with varying levels of permissions and restrictions (e.g., CC BY, CC BY-SA, CC BY-NC).
 - ii. License terms can include attribution, non-commercial use, share-alike requirements, and more.
 - iii. Promote sharing and collaboration while respecting the creator's intentions.
- Creative Commons licenses are used for a wide range of creative content, including educational materials, photographs, music, and written works.